

## **Algorithms and Data Structures Narrative**

This artifact is a backend Contact Service that would likely be implanted in frontend functionality. Within it there are two testing files that test the Contact class, and the ContactService itself. This artifact was created in May 2025 in one of my previous classes. I selected this item because it demonstrates working with data structures, as well as testing implementation which is an important aspect in secure coding. I improved this artifact by porting it over from Java to Python. Doing this required a few changes to the core project, one of which was using pytest for testing rather than JUnit tests. I also changed the data structure being used from an array to a dictionary because I felt that this fits better with the overall program considering each contact would have a unique contact ID. This makes the dictionary a superior choice for retrieving data since the contact ID can be used to pull specifically what is being looked for rather than having to scan an array. I did meet my planned course outcome which was course outcome three. This outcome is designing and evaluating computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices. I feel I met this outcome since porting the project over required changes in the testing tool being used, as well as using a superior data structure for the project at hand. The process of completing this enhancement was straightforward once I became familiar with the syntax of pytest compared to JUnit tests. I first ported over the Contact and ContactService files, which was the easier part of the enhancement. From there, I just implemented each test from the Java project one by one into my Python project using pytest syntax. One problem I did encounter was not using pytest as the Python interpreter, which I fixed by going into the settings and manually setting the interpreter to

use pytest. This enhancement helped me learn more about creating unit tests in Python, which I had never done before this.